IAS PRELIMS/MAINS

SCIENCE & TECHNOLOGY

A COMPREHENSIVE COVERAGE

MODULE - 32

INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 13

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

LI-FI TECHNOLOGY

- Li-Fi or Light Fidelity is a wireless technology that makes use of visible light in the place of radio waves to transmit data at terabits per second speed (more than 100 times faster than Wi-Fi).
- Li-Fi offers a great promise to overcome the existing limitations of Wi-Fi by providing for data heavy communication in short ranges.
- Li-Fi is a solution for wireless pollution.
- We can use LED bulbs which consume less power for transmitting data. To make LED lamps as hot spots of data, electrical data signal should be converted into photons (light) at the source and at the receiver device an optical detector is needed which converts light back into electricity.
  - ✓ Wireless Pollution or Electro Pollution is the exposure to large doses of radio waves. In the long term it can cause, leukaemia, other types of cancers and other health problems such as failure of organs etc.

HARALD HAAS

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

LI-FI TECHNOLOGY

- For commercial utilization, Li-Fi must exhibit three ‘Hs’ – High Data Rates, High Reliability and High Affordability.
- Li-Fi is currently being tested for indoor usage, but it is sure to go outdoor in a big way by making use of the existing infrastructure used for street and traffic lights.

POTENTIAL APPLICATIONS OF LI-FI TECHNOLOGY

1 TRAFFIC MANAGEMENT

Traffic lights can communicate with vehicles; vehicles can communicate with each other through their headlights and tail lamps and prevent accidents. Each traffic light and street light can be converted into wireless hot spots.

2 VISIBLE LIGHT BEING SAFER

Li-Fi can be used in places where radio waves cannot be used, such as in petrochemical and nuclear plants and hospitals. Also in aircrafts, where most of the communication is made through radio waves, there are restrictions on passenger communication using the same media. This can be easily handled by Li-Fi.

3 UNDERWATER

Li-Fi can also easily work underwater where Wi-Fi fails completely. This opens endless possibilities for military and navigational operations.

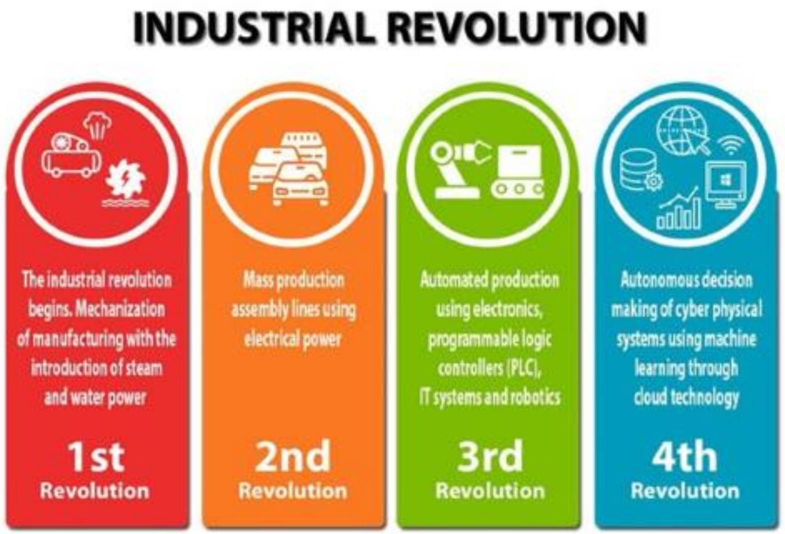
LI-FI TECHNOLOGY

CHALLENGES

- As visual light cannot pass through opaque objects and needs line of sight for communication, the range of Li-Fi is going to be limited.
- Li-Fi is also likely to face interference from external light sources such as sunlight and bulbs or obstructions in the path of transmission and hence interruptions in communication maybe caused.
- Also, initially the installation costs maybe high.

THE FOURTH INDUSTRIAL REVOLUTION (4IR)

- The Fourth Industrial Revolution is the fourth major industrial era since the initial Industrial Revolution of the 18th century. It is marked by emerging technology breakthroughs in a number of fields, including robotics, artificial intelligence, nanotechnology, quantum-computing, biotechnology, the Internet of Things, fifth-generation wireless technologies (5G), additive manufacturing/3D printing and fully autonomous vehicles.
- These technologies are disrupting almost every industry in every country. And the breadth and depth of these changes started the transformation of entire systems of production, management, and governance.



FIRST INDUSTRIAL REVOLUTION

- The First Industrial Revolution took place from the 18^{th} to 19^{th} centuries in Europe and North America. It was a period when mostly agrarian, rural societies became industrial and urban.
- The iron and textile industries, along with the development of the water wheel and then the steam engine, played central roles in this period.

SECOND INDUSTRIAL REVOLUTION

- The Second Industrial Revolution took place between 1870 and 1914, just before World War I. It was a period of growth for pre-existing industries and expansion of new ones, such as steel, oil and electricity, and used electric power to create mass production.
- Major technological advances during this period included the telephone, light bulb, phonograph and the internal combustion engine.

THIRD INDUSTRIAL REVOLUTION

- The Third Industrial Revolution, or the **Digital Revolution**, refers to the advancement of technology from analog electronic and mechanical devices to the digital technology available today.
- The era started during the 1980s. Advancements during the Third Industrial Revolution include the personal computer, the internet, and information and communications technology (ICT).

FOURTH INDUSTRIAL REVOLUTION

- The Fourth Industrial Revolution builds on the Digital Revolution, representing new ways in which technology becomes embedded within societies and even the human body.
- The biggest impact of the Fourth Industrial Revolution will be improving the quality of life, reducing inequality of the world's population and raising the income level.

HOW IS THE FOURTH INDUSTRIAL REVOLUTION FUNDAMENTALLY DIFFERENT FROM THE PREVIOUS THREE?

- The previous three were characterized mainly by advances in technology.
- The underlying basis for 4IR lies in advances in communication and connectivity rather than technology.
- These technologies have great potential to continue to connect billions of more people to the web, drastically improve the efficiency of business and organizations and help regenerate the natural environment through better resources management.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by

Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

For Information:
info.learningspacedigital@gmail.com

Visit us at:
www.learningspacedigital.com
98499 42299